

Alireza Nezamzadeh

Curriculum Vitae

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🌐 [Alireza Nezamzadeh](#)

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EDUCATION

M.Sc. Electrical Engineering-Control

Isfahan University of Technology (IUT)

Sep. 2019 – Feb. 2022

Thesis: "Path Planning for single agent guidance and environment coverage by two agents in the discrete environment using reinforcement learning"

Supervisor: Assoc.Prof. Marzieh Kamali

GPA: 18.08/20.00 (4.00/4.00)

B.Sc. Electrical Engineering-Power

Shahrekord University (SKU)

Sep. 2015 – July. 2019

Thesis: "Design and construction of single-phase inverter for photovoltaic system"

Supervisor: Assoc.Prof. Navidreza Abjadi

GPA: 15.92/20.00 (3.27/4.00)

RESEARCH INTERESTS

- Control systems
- Learning-based controllers
- Reinforcement Learning
- Autonomous Vehicles
- Multi-agent system
- Path Planning

AWARDS & HONORS

- 🏆 **Ranked third** among 15 M.Sc. students of Control engineering, Isfahan University of Technology, Isfahan, Iran, 2022
- 🏆 **Ranked first** among 24 B.Sc. students of Electrical Power Engineering, Shahrekord University, Shahrekord, Iran, 2019
- 🏆 **Third Place** at 30th Electrical Safety Conference – Esfahan Regional Electricity Company (EREC), Isfahan, Iran, 2023
- 🏆 **Talented Student Award:** Qualified for graduate studies at Isfahan University of Technology without an entrance exam 2019

PUBLICATION

- **A. Nezamzadeh**, H. Jalaly Bidgoly, and M. Kamali, "Cooperative Coverage Path Planning Using Q-Learning and Sarsa in Two Environments", In 2024 32nd International Conference on Electrical Engineering (ICEE), pp. 1-5. IEEE, 2024. [\[Link\]](#)
- **A. Nezamzadeh**, M. Esmailidehkordi, "Disturbance Rejection in Quadruple-Tank System by Proposing New Method in Reinforcement Learning", In 2024 14th International Conference on Computer and Knowledge Engineering (ICCKE), pp. 137-142. IEEE, 2024. [\[Link\]](#)
- **A. Nezamzadeh**, M. Esmailidehkordi, "Fault Tolerant Control of Synchronous Generator Using a New MPC Controller Approach", In 29th International Electrical Power Distribution Conference (EPDC), [Accepted](#).
- M. Esmailidehkordi, **A. Nezamzadeh**, M. Zekri, I. Izadi, F. Sheikholeslam, "Type-2 Fuzzy Wavelet Control for a Quadruple-Tank System based on Disturbance Rejection", In 33rd International Conference on Electrical Engineering (ICEE), [Accepted](#).
- M. Esmailidehkordi, M. Zekri, I. Izadi, F. Sheikholeslam, **A. Nezamzadeh**, W. Kinsner, S. Hosseinpour, N. Sepehri, "Sequential Type-2 Fuzzy Wavelet Extreme Learning Machine for Robust Online Control Applications", [Submitted to IEEE Transactions on Control Systems Technology](#).
- **A. Nezamzadeh**, M. Esmailidehkordi, A. Yazdani, H. Habibi "Rapid Fault Detection and Fault-Tolerant Control of Permanent Magnet Linear Motors in the Presence of Actuator Faults, Disturbance, and Sensor Noise", [Preprint journal article](#).

- **A. Nezamzadeh**, M. Esmailidehkordi, M. Naghdi, I. Izadi” A Hybrid Brain Emotional Learning Controller Combined with a Fuzzy Extended State Observer for Robust Control of Piezoelectric Actuators”, [Preprint journal article](#).
- **A. Nezamzadeh**, M. Esmailidehkordi, ”Rapid Fault Detection Strategy for Synchronous Generators Connected to Power Grid”, In 2024 14th Smart Grid Conference (SGC), [Submitted](#).

CERTIFICATES

PLC Programming

Aug. 2016

- Isfahan Province Technical and Vocational

LOGO Programming

Aug. 2016

- Isfahan Province Technical and Vocational

Matlab Programming

Jan. 2019

- Electrical Academic Society of Shahrekord University

ACADEMIC PROJECTS

Nonlinear Real-Time Systems

Nov. 2023 – Present

- Robust control of [the Inverted Pendulum](#) with unknown parameters
- Robust control of [Twin Rotor MIMO](#) system with external disturbances
- Control of [Piezoelectric Actuators](#) without hysteresis model using adaptive brain emotional learning-based Intelligent controller (BELBIC)

Multi-Agent Systems

Sep. 2020 – Feb. 2022

- Cooperative Path Planning Using Reinforcement Learning
- Cooperative Control of Multi-Agent Systems
- Formation Control and Collision Avoidance

Learning Algorithms

Sep. 2020 – Feb. 2022

- Path Planning for Single Agent Guidance in a Discrete Environment Using Reinforcement Learning
- Environment Coverage by Two Agents in a Discrete Environment Using Reinforcement Learning
- Increasing The Convergence Speed of Q-Learning and SARSA Learning Algorithms

Control of a mobile robot

Sep. 2020 – Feb. 2022

WORK EXPERIENCE

ACADEMIC

- **Research Assistant (RA) – “My Current Role”**

July. 2024 – Present

Remote Researcher, Murdoch University

International Collaborator: [Dr. Amirmehdi Yazdani](#)

Nov. 2023 – Present

Nonlinear Control Lab, Isfahan University of Technology

Co-authors: [Dr. Marzieh Kamali](#) and [Dr. Iman Izadi](#)

- **Research Assistant (RA)**

Sep. 2020 – Feb. 2022

Nonlinear Control Lab, Isfahan University of Technology

Supervisor: [Dr. Marzieh Kamali](#)

- **Teacher Assistant (TA)**

Mar. 2020 – Feb. 2022

- 1) Adaptive Control Systems
- 2) Linear Control Systems
- 3) Artificial Neural Networks

Supervisor: [Dr. Marzieh Kamali](#)

INDUSTRY

- **Electrical Engineer**

Isfahan's Mobarakeh Steel Company, Isfahan, Iran

Jul. 2018 – Oct. 2018

Short-term internship to get familiar with the working environment

- **Electrical Engineer**

Esfahan Regional Electrical Company (EREC), Isfahan, Iran

Aug. 2022 – Oct. 2023

Alternative civilian service

SKILLS

Programming: MATLAB, Python

Microcontrollers: AVR

Software: Simulink, Google Colab, Proteus, Jupyter Notebook

LANGUAGES

English: IELTS Score: 6.5 (Listening: 7, Reading: 6, Speaking: 7, Writing: 6)

Persian: Native

RELEVANT COURSES

- 1) **Neural Networks (MSc): 18.96/20 - Instructor:** Assoc.Prof. Marzieh Kamali
- 2) **Systems Identification (MSc): 19.5/20 - Instructor:** Assoc.Prof. Maryam Zekri, Top Score in The Class
- 3) **Optimal Control Systems (MSc): 18.75/20 - Instructor:** Assoc.Prof. Javad Askari Marnani
- 4) **Multivariable Control Systems (MSc): 17.5/20 - Instructor:** Assoc.Prof. Mohsen Mojiri, Top Score in The Class
- 5) **Adaptive Control (MSc): 17/20 - Instructor:** Assoc.Prof. Marzieh Kamali
- 6) **Predictive Control (MSc): 17/20 - Instructor:** Dr. Yadollah Zakeri

REFERENCES

- [Dr. Marzieh Kamali](#)

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Department of Electrical Engineering
Isfahan University of Technology
Email: m.kamali@iut.ac.ir

- [Dr. Hamed Jalaly Bidgoly](#)

Assistant Professor of Electrical Engineering
Department of Electrical Engineering
Isfahan University of Technology
Email: jalaly.hamed@iut.ac.ir

- [Dr. Majdeddin Najafi](#)

Assistant Professor of Electrical Engineering
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